RESEARCH PROPOSAL

Quantum Computing vs. Traditional Predictions/AI for Stock Portfolio Selection



Can simulated quantum algorithms help build better investment portfolios?

Zachariah Seaford • CECS 385 • October 2026

The problem: Promise vs Evidence

The opportunity

There are many ways to combine stocks, and finding a good portfolio gets harder as the options grow.

The uncertainty

Quantum computing could offer another way to solve difficult investment problems.

Who is affected

Investors, financial advisers, and firms looking for better ways to balance risk and returns.

Sources: [1], [2]

Research question & hypotheses

Can a simulated quantum computing algorithm produce better risk-adjusted portfolio returns than traditional and AI-assisted investment methods after trading costs?

H_0 · Null hypothesis

Quantum computing will not pick better investments than traditional computers or AI.

H_1 · Alternative hypothesis

Quantum computing will pick better investments than traditional computers or AI.

Quantum vs. Classical Computers vs. AI

Traditional computing

BITS • 0s AND 1s



Runs regular investment models and checks possible portfolio choices.

Quantum computing

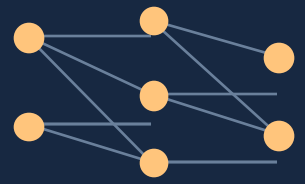
QUBITS • QUANTUM OPERATIONS



Uses a different way to search for good solutions to optimization problems.

Artificial intelligence

AI • LEARNS FROM DATA



Studies historical market patterns to help predict future returns.

KEY TERMS

QAOA
Quantum Approximate Optimization Algorithm

ETF
Exchange-Traded Fund

Qubit
Quantum bit

Qiskit
IBM's quantum software toolkit that will be loaded into a Python library

Sharpe ratio
Return compared with risk

Study design: a focused comparison

Data

Proposed sample: six liquid U.S. equity-sector ETFs.
Daily adjusted prices, 2015–2025.

Estimate returns and correlations using past data only.

Portfolio task

Choose exactly three of six ETFs. Hold selected ETFs at equal weights.

Balance estimated return against portfolio risk.

Methods compared

Quantum: Quantum Approximate Optimization Algorithm on a simulator.

Classical: exhaustive-search optimum and a simple greedy heuristic.

Data-Set

20 year ETF analysis dataset: https://github.com/manisahnhttps://github.com/manisahni/marketdatai/marketdata

Methods informed by: [1], [2]

How I will simulate quantum computing

I will use regular computer software to simulate a small quantum portfolio-selection algorithm.

STEP 01

Collect market data

Gather historical ETF prices, returns, and risk estimates.

→

STEP 02

Build the portfolio problem

Turn six ETFs into six yes/no choices. Select exactly three.

→

STEP 03

Run the quantum simulation

Use Python + Qiskit to simulate QAOA.

→

STEP 04

Record and compare results

Save the selected ETFs, score, runtime, and settings.

How I will compare the models

Quantum model

QAOA simulator



Using Qiskit in python to search for a strong combination of three ETFs with a simulated quantum circuit.

AI model

Random Forest



Uses historical market data to predict returns and help select investments.

Classical model

Exhaustive search

20

Checks all 20 possible three-ETF portfolios to find the highest-scoring one.

Fair comparison: same historical test periods, portfolio rules, and trading costs.

Validation: avoid an impressive but fragile back-test



Learn from historical prices

Apply quantum and classical portfolio selection

Hold for next month; deduct trading costs

Measure returns and risk

Backtesting cautions: [3]. Proposed cost assumption: 10 basis points per trade; sensitivity checks at 0 and 25 bps.

Ethical considerations

Transparency

Publish data rules, model settings, costs and code; do not equate back-tests with real investment returns.

Fairness & access

Discuss whether proprietary quantum resources could increase differences between well-funded and smaller investors.

Accountability

No automated real-money trades; human review of conclusions and explicit uncertainty statements.

Data & research ethics

Use public market data. No human subjects planned; an IRB review may be needed only if interviews are added.

Responsible innovation and human-subjects guidance: [4], [5]

Expected findings & contribution

What I expect to happen: Quantum optimization may produce competitive portfolios, but I don't expect to demonstrate a clear advantage at such a small scale.

If the quantum method performs better

I'll document the size of the improvement and whether it continues after trading costs.

If it does not

I'll report that result too. Understanding when quantum methods don't help is valuable research.

Study limitations

Simulation ≠ real hardware

A noiseless or idealized simulator does not reproduce device noise, queueing or all runtime costs.

Tiny asset universe

Six ETFs and fixed equal weights may not generalize to institutional portfolios.

Historical data only

Back test results can change in new regimes; price history misses real-world market frictions.

Not a speedup proof

A small Quantum Approximate Optimization Algorithm simulation cannot establish a scalable quantum computational advantage.

References

[1] R. Orús, S. Mugel, and E. Lizaso, “Quantum computing for finance: Overview and prospects,” Reviews in Physics, vol. 4, art. 100028, 2019. doi: 10.1016/j.revip.2019.100028.

[2] D. Herman et al., “Quantum computing for finance,” Nature Reviews Physics, vol. 5, pp. 450–465, 2023. doi: 10.1038/s42254-023-00603-1.

[3] D. H. Bailey, J. M. Borwein, M. López de Prado, and Q. J. Zhu, “The probability of backtest overfitting,” Journal of Computational Finance, vol. 20, no. 4, 2017. doi: 10.21314/JCF.2016.322.

[4] J. Stilgoe, R. Owen, and P. Macnaghten, “Developing a framework for responsible innovation,” Research Policy, vol. 42, no. 9, pp. 1568–1580, 2013. doi: 10.1016/j.respol.2013.05.008.

[5] National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research, The Belmont Report, 1979.

References are numbered in the order cited on the slides.